Thuan Phuoc Nguyen

PhD Candidate in Robotics | **Robotics / AI Internship**

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Project videos

PhD candidate at Aalto University researching 3D scene graphs for long-term scene understanding and robot autonomy. Experienced in translating research problems into working robotic systems, both independently and collaboratively. Published in IEEE RA-L and IROS. Available for internship from June 2027; open to earlier research collaboration.

SELECTED EXPERIENCE

Visiting Researcher • University of Freiburg, Freiburg im Breisgau, Germany — Oct 2026 – Present

- Research long-horizon mobile manipulation task planning using scene graphs.

Robotics Engineer, Part-Time • Skya, Turku, Finland — Feb 2026 – Sep 2026

- Built a smart landing pad prototype using ultra-wideband UAV localization and 3D LiDAR obstacle detection.

Doctoral Researcher • Aalto University, Espoo, Finland — Oct 2023 – Present

- Research and develop 3D scene graph methods for long-term semantic scene understanding.
- Supervise student projects on mobile manipulation, object detection/tracking, and long-term autonomy.

Research Assistant / Project Researcher • University of Turku, Turku, Finland — Feb 2022 – Aug 2023

- Conducted applied research on real-time, autonomous safe UAV landing with panoramic sensor for master's thesis.

SELECTED AI/ROBOTICS PROJECTS

EGG: Event-Grounding Graphs for Robot Observations — Apr 2025 – Mar 2026

Python, ROS, LLMs, VLMs | github.com/aalto-intelligent-robotics/EGG

- Designed a spatio-temporal scene graph framework linking robot-observed human events with relevant objects; integrated VLM event captioning and LLM-based question answering.

REACT: Real-Time Updatable 3D Scene Graphs — Oct 2024 – Mar 2025

Python, C++, PyTorch, ROS | github.com/aalto-intelligent-robotics/REACT

- Extended Hydra 3D scene graph framework and trained an object reID model for real-time scene graph updates.

Autonomous UAV Exploration & Mapping (1st place at ENRICH Robotic Hackathon 2023) — Feb 2023 – Jun 2023

Python, C++, ROS, PX4, Gazebo

- Developed a ROS/PX4 UAV system for frontier-based exploration, 3D mapping, and LiDAR obstacle avoidance.

Smart UAV Landing Pad with Panoramic Vision — Oct 2022 – Apr 2023

Python, ROS, PX4, Gazebo | github.com/phuoc101/safe_landing

- Fine-tuned and deployed a YOLO human detector on a Jetson Xavier NX for panoramic landing-area monitoring and safety-aware UAV hovering/repositioning.

Data-Driven Model Learning for Robot Simulation and Control — Aug 2022 – Jan 2023

Python, PyTorch, ROS, Gazebo | github.com/phuoc101/GP_RL

- Controlled a real Avant loader manipulator using Gaussian process modeling and reinforcement learning.

SELECTED PUBLICATIONS

- P. Nguyen, F. Verdoja, and V. Kyrki, "Event-grounding graph: Unified spatio-temporal scene graph from robotic observations," *IEEE Robotics and Automation Letters*, 2026.
- P. Nguyen, F. Verdoja, and V. Kyrki, "REACT: Real-time Efficient Attribute Clustering and Transfer for Updatable 3D Scene Graph," *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025.
- P. T. Nguyen, T. Westerlund, and J. Peña Queralta, "Vision-based safe autonomous UAV docking with panoramic sensors," *Frontiers in Robotics and AI*, 2023.

TECHNICAL SKILLS

Programming: Python, C++

Robotics & AI: ROS, Gazebo, PX4, PyTorch, OpenCV

Tools: Docker, Linux, Git, GitHub, CVAT

EDUCATION

Tampere University — Tampere, Finland

MSc.Tech, Robotics and Artificial Intelligence — *Aug 2021 – Jun 2023*

Ho Chi Minh City University of Technology — Ho Chi Minh City, Vietnam

B.Eng., Mechatronic Engineering — *Aug 2015 – Nov 2020*